Predicting Fatal vs. Non-Fatal Gun Violence Incidents from Structured Incident Records

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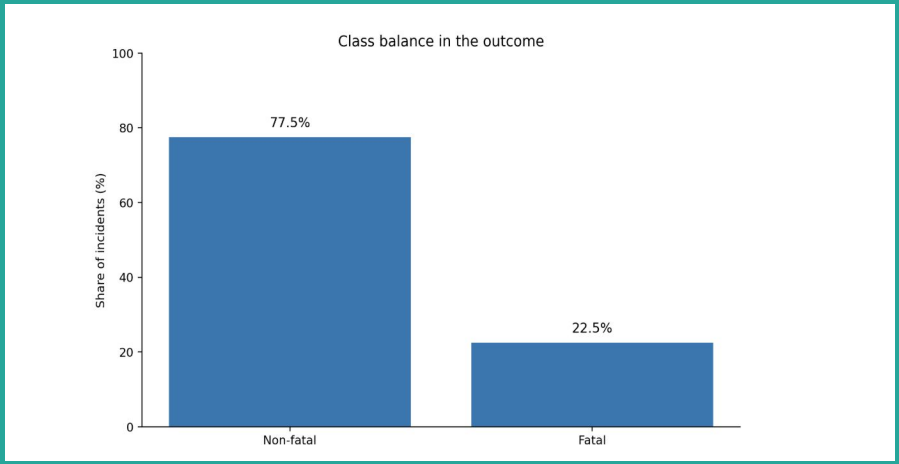
- Problem:
  - Structured incident records are widely available, but it is unclear whether they contain enough defensible information to support meaningful severity screening once outcome-adjacent and documentation-heavy fields are removed.
- Research Question:
  - Do structured incident-level records still contain usable signal for fatal-vs-non-fatal screening, and does that signal come mainly from immediate incident descriptors or from broader contextual markers such as time and place?

Predicting Fatal vs. Non-Fatal Gun Violence Incidents from Structured Incident Records

- Why this matters
  - Gun violence is a major public safety issue, and incidents differ meaningfully in severity.
  - Structured incident records are widely available, so it is important to understand whether they can support meaningful severity screening.
  - This question becomes more important once less defensible variables are removed, because it is no longer obvious how much predictive information remains.
- Hypothesis
  - I expect structured incident records to retain meaningful predictive signal even under a more defensible feature design.
  - I also expect model choice and decision-rule choice to matter for practical performance.

Data and Outcome

- Data
  - Source: Gun Violence Archive incident listings
  - Analysis starts from the saved cleaned file stage3.csv
  - 239,677 incidents
  - Coverage: 2013-01-01 to 2018-03-31

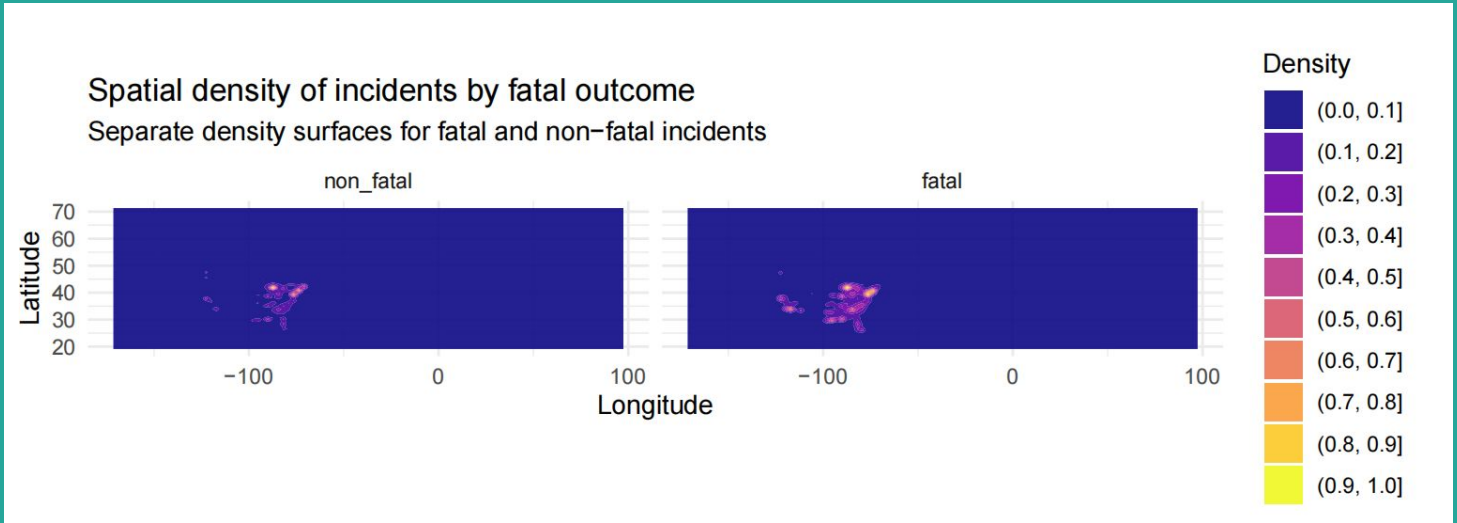


- Outcome
  - fatal = 1(n_killed > 0)
  - Fatal share: 22.5%

- Why this matters for modeling
  - This is an imbalanced classification task, so raw accuracy alone is not informative.

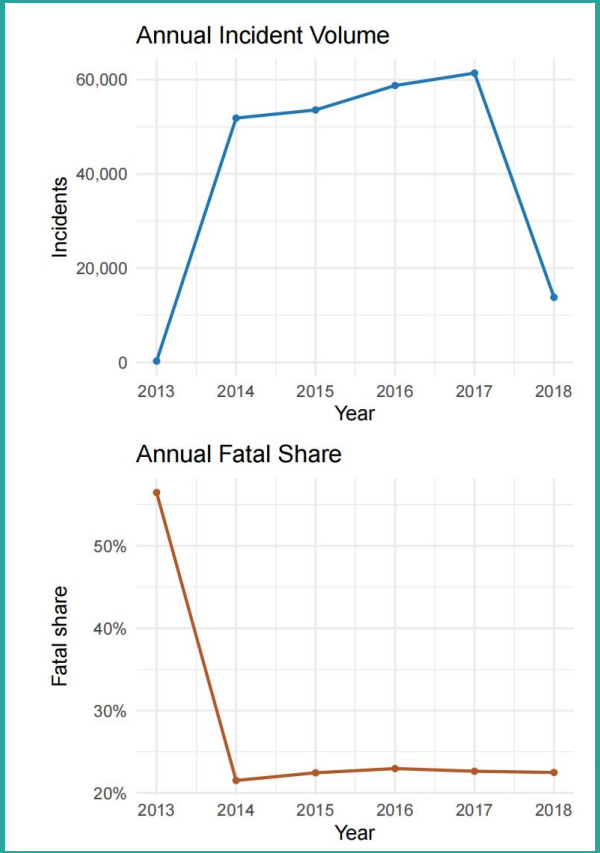
Minimal EDA: Spatial Pattern

- Incident records show clear spatial clustering rather than an even distribution across space.
- Fatal and non-fatal density patterns overlap only partly, not completely.
- This suggests that some usable signal may be contextual or place-related rather than strongly incident-specific.



Minimal EDA: Time Pattern

- Incident volume and fatal share show different time patterns.
- Higher incident counts do not necessarily imply a higher fatal share.
- This suggests that the prediction task is not driven by year-level trend alone.
- Later models therefore need to rely on broader structured and contextual information.



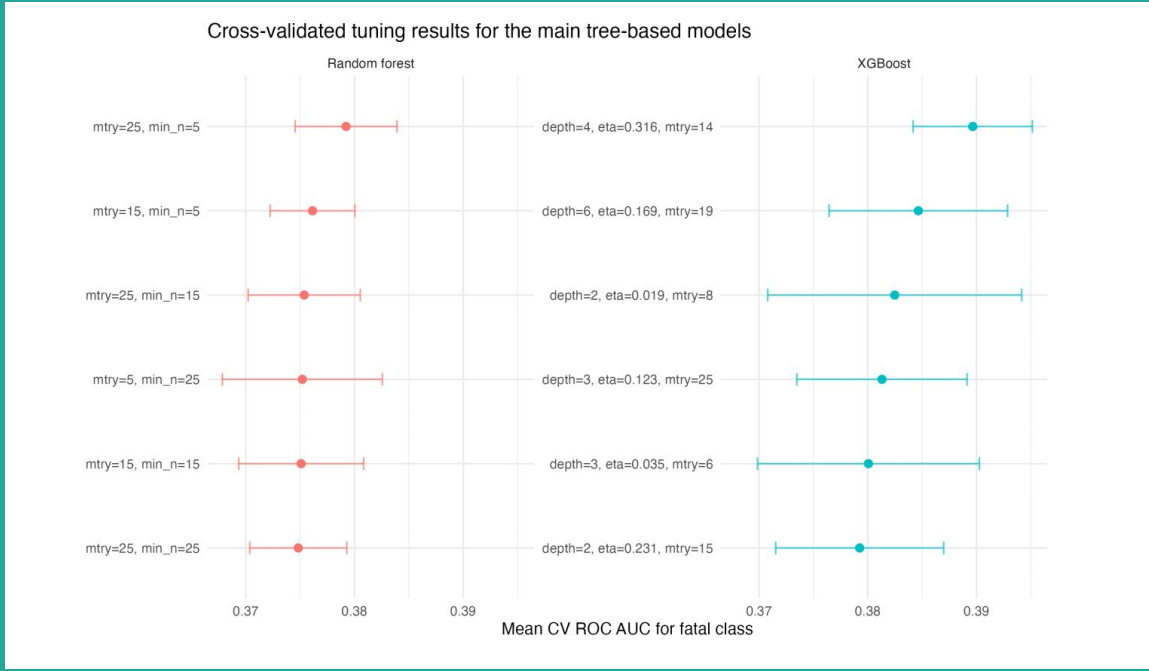
Baseline Models and Main Evaluation Results

- Models compared
  - Logistic regression
  - Random forest
  - XGBoost
- Evaluation focus
  - Held-out ROC AUC for underlying classifier quality
  - Fatal sensitivity at the default 0.50 cutoff
- Main baseline result
  - Random forest is the strongest underlying classifier.
  - But all three baseline rules perform poorly for fatal-case detection.

model	roc_auc	accu-racy	bal_accu-racy	sensitivity_fa-tal	specificity_nonfa-tal
Random forest	0.6632	0.7738	0.5233	0.0684	0.9782
Logistic regression	0.6348	0.7721	0.5165	0.0523	0.9807
XGBoost	0.6234	0.7753	0.5000	0.0000	1.0000

Tuning and Leakage-Safe Model Selection

- I split the data first and fit all learned preprocessing only on the training workflow.
- Tree-based models were tuned within the training/validation pipeline.
- This makes the baseline comparison more defensible and avoids leakage.
- But tuning alone does not solve the main practical problem: default classification remains too conservative under class imbalance.



Class Imbalance and Practical Decision Rules

- Why imbalance matters
  - Fatal incidents are the minority class.
  - The default 0.50 cutoff misses too many fatal cases.
  - So the final task is not only choosing the strongest classifier, but choosing a practical decision rule.

- Methods considered
  - Class weighting
  - Oversampling
  - Threshold moving

- Key idea
  - Different imbalance strategies reflect different trade-offs between fatal-case detection and false positives.

Final Results and Recommended Operating Rule

held-out test performance of thresholds selected on the validation split under a specificity constraint of about 0.75

model	threshold	bal_accuracy	sensitivity	specificity
Random forest	0.27	0.602	0.462	0.743
XGBoost	0.25	0.551	0.238	0.864
Logistic regression	0.28	0.579	0.400	0.757

- Random forest gives the strongest overall compromise.
- Logistic regression remains a competitive and simpler benchmark.
- XGBoost becomes more conservative under the specificity-constrained rule.

What We Learn and Contribution

- Structured incident-level records retain limited but still useful predictive signal under a more defensible feature design.
- The task is better understood as screening than as strong event-level classification.
- Much of the usable signal appears contextual, especially time- and place-related, rather than driven by one strong incident-specific predictor.
- The final recommended model is a specificity-constrained threshold-moving random forest.
- This final model should be understood as a practical operating-point choice for this dataset and use case, not a universal solution or a causal finding.

Thank you